NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER 1 QUESTIONS AND SOLUTIONS.

SECTION A: 5 marks each

1. Find the value of y if $x + y = 8$ and $\log_2(xy) = 2$.

Solution

From $x + y = 8 \rightarrow y = 8 - x$ thus $\log_2(x(8-x)) = 2 \Leftrightarrow x^2 - 8x + 4 = 0$ whose solutions are $x = 4 \pm 2\sqrt{3}$ thus $y = 8 - (4 \pm 2\sqrt{3}) = 4 \pm 2\sqrt{3}$.

2. Let the terms $a_1, a_2, a_3, \ldots$ be a sequence such that first term $a_1 = 2023$ and each succeeding term in the sequence is obtained as the sum of the cubes of the digits of the previous term. What is the value of the term a_{2024}?

Solution

$a_1 = 2023, a_2 = 2^3 + 0^3 + 2^3 + 3^3 = 43, a_3 = 4^3 + 3^3 = 91, a_4 = 9^3 + 1^1 = 730, a_5 = 7^3 + 3^3 + 0^3 = 370, a_6 = 3^3 + 7^3 + 0^3 = 370$. The terms in the sequence become constant as $n \rightarrow \infty$ for $n \geq 5$ thus $a_{2024} = 370$.

3. Solve for x in the equation $6^{\frac{1}{x}} + 3 \cdot 9^{\frac{1}{x}} = 2 \cdot 4^{\frac{1}{x}}$.

Solution

Re-write $6^{\frac{1}{x}}$ as $2^{\frac{1}{x}} \cdot 3^{\frac{1}{x}}$, $9^{\frac{1}{x}}$ as $3^{\frac{2}{x}}$ and $4^{\frac{1}{x}}$ as $2^{\frac{2}{x}}$ subsituting these into the main equation we get, $2^{\frac{1}{x}} \cdot 3^{\frac{1}{x}} + 3 \cdot 3^{\frac{2}{x}} = 2 \cdot 2^{\frac{2}{x}}$ dividing through by $3^{\frac{1}{x}}$ and then 2 we get, $1 + 3 \cdot \left(\frac{3}{2}\right)^{\frac{1}{x}} = 2 \cdot \left(\frac{2}{3}\right)^{\frac{1}{x}}$, let $y = \left(\frac{2}{3}\right)^{\frac{1}{x}}$ then the equation becomes $1 + 3y^{-1} = 2y \Leftrightarrow 2y^2 - y - 3 = 0$ whose solution is either $y = \frac{3}{2} \Rightarrow x = -1$ or $y = -1 \Rightarrow \left(\frac{2}{3}\right)^{\frac{1}{x}} = -1$ which has no solution. Thus the only solution is $x = -1$.

4. For all values of x, the function $f(x)$ satisfies $f(x+1) + f(x-1) = f(x)$. If $f(0) = 5$ and $f(1) = 7$, then find the value of $f(2022) + f(2023) + f(2024)$.

Solution

$f(x+1) = f(x) - f(x-1) \Rightarrow f(x+2) = f(x+1) - f(x)$ and $f(x+1) = f(x) - f(x-1)$, from these we have $f(x+2) = -f(x-1) \rightarrow f(x+3) = -f(x)$ and $f(x+6) = -f(x+3)$ but $f(x+3) = -f(x)$ thus $f(x+6) = f(x)$ which implies that the function is periodic with period 6; the following is true about the functional values of $f(x)$, $f(2022) = f(0) = 5$, $f(2023) = f(1) = 7, f(2024) = f(2) = 2$, since $f(2) = f(1) - f(0) = 7 - 5 = 2$ Thus $f(2022) + f(2023) + f(2024) = 5 + 7 + 2 = 14$.

5. There are 120 different 5-digit numbers that use each of the digits $2, 4, 5, 7$, and 8 exactly once. If listed in numerical order from smallest to largest, what number is in the 97th position of the list?

Solution

There are 24 numbers that start with each digit so the first 96 numbers consist of all the numbers which start with $2, 4, 5$ and 7 therefore the 97th number in the list is the smallest which starts with 8 that is $82,457$.

6. Suppose that increasing the bus fare by 10% results in a 5% decrease in the number of passengers. To achieve an 8% increase in revenue, what percentage increase in the fare should be implemented? Find the required fare increase percentage assuming that the fare increase does not exceed 50%.

Solution

Let c_0 and n represent the original cost and the number of passengers respectively, the cost and the number of passengers after the increase are $c_0\left(1+\frac{x}{100}\right)$ and $n\left(1-\frac{x}{200}\right)$. The revenue becomes $c_0\left(1+\frac{x}{100}\right)\cdot n\left(1-\frac{x}{200}\right) = nc_0\left(1+\frac{8}{100}\right)$ solving gives $x^2-100x+1600=0 \Leftrightarrow x = 20\%$ or $x = 80\%$ thus $x = 20\%$ as $x \leq 50$.

7. A worker can finish a job in 3 days with his present supply of a certain type of machine. However, with 3 more of these machines the job could be done in 2 days. If the machines all work at the same rate, how many days would it take to do the job with one machine?

Solution

Let x be the number of machines that can finish the job in 3 days then $3x = 2(x+3)$ is the number of days that one machine would take to finish the job thus $3 \times 6 = 18$ days.

8. Four suspects of a crime made the following statements to the police:

Ryan said "Bruce did it."
Andrew said "I didn't do it."
Bruce said "Anne did it."
Anne said: "Bruce lied when he said I did it."

Determine who committed the crime if exactly one of the four statements is

(a) true

(b) false.

Solution

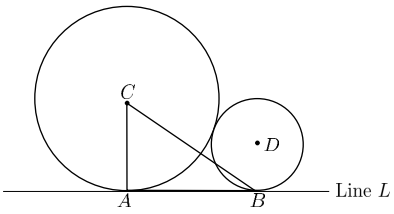
Both answers in (a) and (b) are obtained by using that each person is the criminal and seeing how many of the statements are true or false in each case. Only Andrew in (a) satisfy the information given. and in (b), it is Bruce.

9. Let A and B be two cars initially positioned at opposite points along a straight Road. When they both travel at their original constant speeds, it takes 30 minutes for them to meet each other. If car A doubles its speed while B maintains the original speed, they meet in 25 minutes. Determine the time in minutes it will take for them to meet if car B doubles its speed while A retains the original speed.

Solution

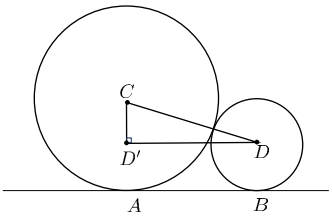
Let the distance between A and B be L then let s_1 and s_2 be the original speed of A and B respectively from the condition we have $(s_1 + s_2)30 = L$ and $(2s_1 + s_2)25 = L$ solving gives $s_1 = L\left(\frac{1}{25} - \frac{1}{30}\right)$ and $s_2 = L\left(\frac{2}{30} - \frac{1}{25}\right)$, we want to find t such that $(s_1+2s_2)t = L$ thus $t = \frac{100}{6} = \frac{50}{3}$ minutes.

10. The larger circle is of radius $7cm$ centered at C is tangent to the smaller circle of radius $1cm$ centered at D. If the circles are tangent to the line L at A and B. What is the area of the triangle ABC?



Solution

Apply the pythogras Theorem to the triangle $CD'D$ as in the diagram below



to get $(CD)^2 = (CD')^2 + (D'D)^2 \Rightarrow CD = \sqrt{(7-1)^2 + (7+1)^2}$ and $CD = 10cm$, so the area of triangle ABC is $\frac{1}{2} \times 8 \times 7 = 28cm$.

SECTION B: 10 marks each

11. The average age of three siblings, Alice, Bob and John is x. It will become $2x$ when John gets as old as Bob is now. It will become $1.5x$ when Bob gets as old as Alice is now. It will become 30 when John gets as old as Alice is now. Find the current age of John.

Solution

Let a represent the age of Alice now, b represent the age of Bob now and J represent the age of John now, thus $\frac{a+b+J}{3} = x \Leftrightarrow a+b+J = 3x$. After n years John will be $J+n = b$ thus new average is $\frac{a+b+J}{3}+n = 2x = x+x \Leftrightarrow x = n = b-J$. After k years $\frac{a+b+J}{3}+k = 1.5x = 0.5x+x$ so $a-b = 0.5x$. Now after y years average is 30 thus $\frac{a+b+J}{3} + y = 30$ from the facts that $y = a - J, x = n = b - J$ and $a - b = 0.5x$ we have $1.5x = a - J$ thus $x + 1.5x = 2.5x = 30$ thus $x = 12$. so the ages are $J, J + 12, J + 18$ with average 12 thus $J = 2, 14$ and 20 years meaning John is 2 years now.

12. Four sparrows, Pip, Pap, Pep and Pop found a dish of seeds that was free of dirt said Pip: "In turns each will take two grains, Plus a third of what remains and I'll go first, then Pep, then Pop, With Pap the last. And then we stop." But Pap cried out, "It isn't fair. Mine's two seeds less than half Pep's share." But Pip was boss; his word was law So little Pap got nothing more. Poor Pap, his share was rather small! How many seeds were there in all?

Solution

Let N be the number of seeds in the dish, the table below shows the amount of seeds each bird takes,

Pip	Remaining	Pep	Remaining
$\frac{1}{3}(N-2)+2$ $=\frac{N+4}{3}$	$N-\left(\frac{N+4}{3}\right)$ $=\frac{2N-4}{3}$	$\frac{1}{3}\left(\frac{2N-4}{3}\right)+2$ $=\frac{2N+14}{9}$	$\frac{2N-4}{3}-\frac{2N+14}{9}$ $=\frac{4N-20}{9}$

Pop	Remaining	Pap
$\frac{1}{3}\left(\frac{4N-20}{9}-2\right)+2$ $=\frac{4N+16}{27}$	$\frac{4N-20}{9}-\frac{4N+16}{27}$ $=\frac{8N-76}{27}$	$\frac{1}{3}\left(\frac{8N-76}{27}-2\right)+2$ $=\frac{8N+32}{81}$

$\Leftrightarrow \frac{8N+32}{81}=\frac{1}{2}\left(\frac{2N+8}{9}\right)-2 \Leftrightarrow N=158$ seeds.

13. All of John's steps with his left foot are 0.5 metres. His right foot steps are the same, except that every third one is twice as long. He starts with his left foot and alternates, except that after every third left foot step, he takes a step back with his left foot. How many steps, forward and backward, does he take to travel the 19 metres from the bus stop to the Ministry of Silly Walks?

Solution

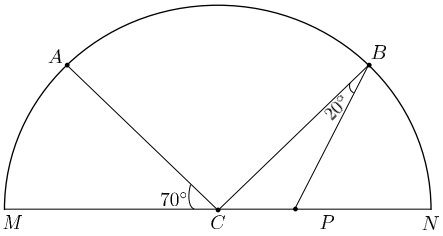
John repeats the sequence $LRLRLBS$ where L and R be the ordinary left and right steps, $B-$ Backward left step and S is a $1m$ right step. This is the 7 steps and the distance covered is $5\times(0.5)-0.5+1=3$. Doing this 6 times will take 42 steps and cover $8m$. John then takes an additional L and an additional R to bring him to the destination thus 44 steps are required.

14. A 5-digit number is said to be "elegant" if (i) its digits are from 1 to 9, (ii) it contains each digit only once, and (iii) it contains the digits $2, 4$ and 8, in that order, but not necessarily consecutively. (For instance, 24385 is elegant). How many elegant numbers are there?

Solution

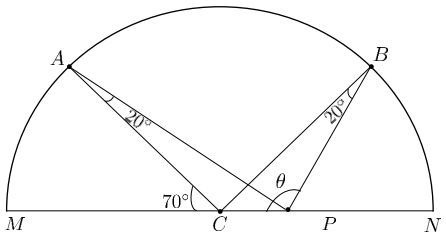
We must select 3 positions for the $2, 4$ and 8 (already chosen order). There are 10 ways to do it namely $248AB, 24A8B, 24AB8$ etc of the remaing 2 positions we have 6 choices for the first digit (anything other than $0, 2, 4, 8$) and 5 for the second (not $0, 2, 4, 8$ or the digit just selected) the total number is $10\times6\times5=300$.

15. Two distinct points A and B are on a semicircle with diameter MN and center C. The point P is on $\overline{CN}$ and $< CBP = 20^\circ$ as indicated in the diagram. What is the value of $< BPN$ if $< ACM = 70^\circ$.



Solution

Consider the figure below



Observe that, $< CPA = 50^\circ$. Let R be the radius of the half cirle and let $\theta =< CPB$ and to $\triangle CPB$ we find that $\frac{\sin 50^\circ}{R} = \frac{\sin 20^\circ}{R}$ and $\frac{\sin \theta^\circ}{R} = \frac{\sin 20^\circ}{CP} \Leftrightarrow \frac{\sin \theta^\circ}{R} = \frac{\sin 50^\circ}{R} \Leftrightarrow \sin\theta = \sin 50^\circ \Rightarrow \theta = 180^\circ - 50^\circ = 130^\circ \Rightarrow < BPN = 180^\circ - 130^\circ = 50^\circ$

END